Monetizing Platforms: An Empirical Analysis of Supply and Demand Responses to Entry Costs in Two-Sided Markets

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Management Science (2025) · Replication

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Goodreads started charging for a free promotion tool in 2018

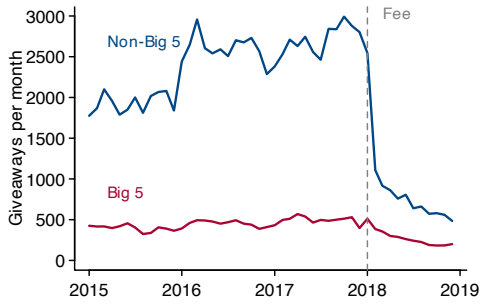
Setting: On Goodreads, authors and publishers give away free books to win reviews and visibility

Shock: $119 per giveaway from 9 January 2018, announced six weeks ahead

Question: How does a fee on publishers affect both publishers and readers?

- Who participates — Table 2
- What gets promoted — Table 3
- How readers respond — Table 4

This replication: Data from the replication files the authors posted at *Management Science*



Big 5: the five largest publishing houses. Giveaways per month, all publishers: about 3,200 in 2017, about 1,150 in 2018

Table 2: 2017–18 vs. 2015–16 (DiD with a historical control group)



$$\underbrace{y_{pt}}_{\substack{\text{giveaways by}\\ p \text{ in month } t}} = \beta_1 \underbrace{\text{Treated}_t}_{\substack{=1 \text{ in 2017}\\ \text{and 2018}}} \times \underbrace{\text{After}_t}_{\substack{=1 \text{ in 2016}\\ \text{and 2018}}} + \beta_2 \text{Treated}_t \times \text{After}_t \times \underbrace{\text{Big5}_p}_{\substack{=1 \text{ for Big 5}\\ \text{publishers}}} + \underbrace{\delta_{m(t)}}_{\substack{\text{month-of-year}\\ \text{fixed effect}}} + \underbrace{\sigma_{p,\text{pair}}}_{\substack{\text{publisher fixed}\\ \text{effect per pair}}} + \varepsilon_{pt}$$

- 14,131 publishers × 48 months, all with a giveaway from 2016 on
- Big 5 = Penguin, HarperCollins, Hachette, Simon & Schuster, Macmillan; the rest: indie and self-publishers

- y: (1) number of giveaways; (2) as a share of its 2015–17 total, which adjusts for publisher size
- β_1: 2018 change, non-Big 5; β_2: extra change, Big 5
- s.e. clustered by publisher

Non-Big 5 publishers cut back twice as much, relative to their size

	(1) Number of giveaways		(2) Proportion	
	Paper	Replication	Paper	Replication
Treated × After	−0.129 (0.023)	−0.129 (0.023)	−0.031 (0.0004)	−0.031 (0.0004)
Treated × After × Big 5	−43.354 (6.561)	−43.354 (6.561)	0.016 (0.002)	0.016 (0.002)
Observations	678,288	678,288	611,376	611,376
R^2	0.883	0.883	0.038	0.038

Paper's conclusion: The fee hits smaller publishers hardest, and the Big 5 gain share

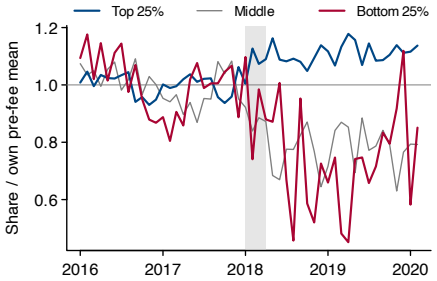
From 2017 to 2018, per publisher and month:

- Non-Big 5: 0.19 → 0.06 giveaways (−68%)
- Big 5: 98.9 → 55.4 giveaways (−44%)
- Scaled by own size (column 2): −0.031 vs. −0.015, about twice the cut

Table 3: popular, middle and niche genres (pre–post regression)

$$\underbrace{\text{Share}_{gt}}_{\substack{\text{genre } g\text{'s share}\\ \text{of giveaway books}\\ \text{in month } t}} = \beta_1 \underbrace{\text{Post}_t}_{\substack{=1 \text{ after}\\ \text{the fee}}} \times \underbrace{\text{Top25}_g}_{\substack{=1 \text{ if top 25\%}\\ \text{(popular)}}} + \beta_2 \text{Post}_t \times \underbrace{\text{Bottom25}_g}_{\substack{=1 \text{ if bottom}\\ 25\% \text{ (niche)}}} + \underbrace{\alpha_g}_{\substack{\text{genre}\\ \text{fixed effect}}} + \underbrace{\delta_{m(t)}}_{\substack{\text{month-of-year}\\ \text{fixed effect}}} + \varepsilon_{gt}$$

- 50 genres × 47 months, Jan 2016 to Feb 2020 without Jan–Mar 2018
- Genre = a book's two most common shelves (tags readers assign, e.g. romance)
- Pre-fee share quartiles: 13 top (popular), 24 middle and 13 bottom (niche) genres
- y: (1) raw share; (2) share relative to the genre's pre-fee mean
- β_1, β_2: change for top, bottom genres
- s.e. clustered by calendar month



Gray band: January–March 2018, dropped

Popular genres gained 11%; niche genres lost 26% of their share

	(1) Raw proportion		(2) Percentage change	
	Paper	Replication	Paper	Replication
Post × Top 25% genre	0.012 (0.001)	0.012 (0.001)	0.110 (0.010)	0.110 (0.010)
Post × Bottom 25% genre	−0.001 (0.0001)	−0.001 (0.0001)	−0.262 (0.039)	−0.262 (0.039)
Observations	2,350	2,350	2,350	2,350
R^2	0.969	0.969	0.188	0.188

Paper's conclusion: Genre diversity falls: popular genres gain share, niche genres lose it

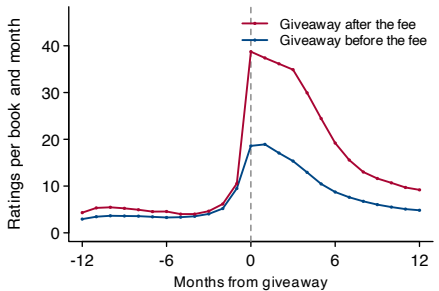
After the fee, each group's share relative to its own pre-fee mean:

- Top 25%, popular (average share 10.5% before): +11%
- Middle 50% (1.6% before): −22%
- Bottom 25%, niche (0.34% before): −26%

Table 4: pre- vs. post-fee cohorts (DiD with two-way fixed effects)

$$\underbrace{N_{jt}}_{\substack{\text{number of ratings of}\\ \text{book } j \text{ in month } t}} = \beta_1 \underbrace{\text{PostGiveaway}_{jt}}_{\substack{=1 \text{ in the 12 months}\\ \text{after the giveaway}}} + \beta_2 \text{PostGiveaway}_{jt} \times \underbrace{\text{PostMon}_j}_{\substack{=1 \text{ if the giveaway}\\ \text{ran after the fee}}} + \underbrace{\alpha_j}_{\substack{\text{book}\\ \text{fixed effect}}} + \underbrace{\gamma_t}_{\substack{\text{year-month}\\ \text{fixed effect}}} + \varepsilon_{jt}$$

- 12 months before and after each giveaway; months without ratings count as 0
- Cohort = when the book's giveaway ran: 58,677 before the fee, 22,931 after
- β_1: giveaway effect, pre-fee cohort; β_2: extra effect, post-fee cohort
- s.e. clustered by book
- Replicated: columns (3) and (4), the number of ratings



Raw means, before fixed effects

After the fee, a giveaway brought three times as many ratings

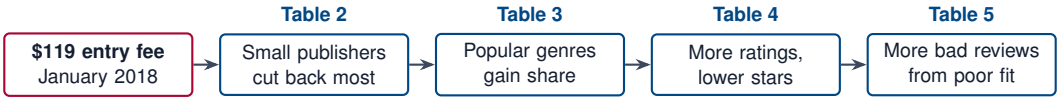
	(3) Number of ratings		(4) Number of ratings	
	Paper	Replication	Paper	Replication
Post-Giveaway	11.008 (0.182)	11.008 (0.182)	6.841 (0.176)	6.841 (0.176)
Post-Giveaway × Post-Mon			14.454 (0.372)	14.454 (0.372)
Observations	2,036,383	2,036,383	2,036,383	2,036,383
Adjusted R^2	0.131	0.131	0.134	0.134

Paper's conclusion: The fee amplifies the giveaway effect: more ratings, but lower scores

After a giveaway, per book and month (sample mean: 9.37 ratings):

- Pre-fee cohort: +6.8 ratings
- Post-fee cohort: 6.8 + 14.5 = +21.3 ratings, about three times as many
- Post-fee cohort's average rating: a further −0.05 stars (column 2, not replicated)

Discussion: one fee, traced across both sides of the market



What the paper adds

- A fee changes *who* stays and *what* is promoted, not just how many join (Armstrong, 2006; Rochet and Tirole, 2003)
- Fixed costs push out niche tastes: preference externalities (Waldfogel, 2003)
- Deals bring more but less satisfied buyers: the Groupon effect (Byers et al., 2012)
- The fee selects on budget, not quality: post-fee books were not rated higher before

How to read the estimates

- The fee hit all at once: Tables 2–3 are before–after changes, net of seasonality
- Table 4 compares cohorts: β_2 mixes which books enter after the fee with how readers respond to them
- All three tables reproduce exactly from the authors' data

Thank you

References

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Replication code and data

GitHub repository

github.com/bobbyZhong10/paper-replication-zhu-ms-2026

`code/`	One R Markdown file per table
`data/`	Data from the authors' replication files
`results/`	Each replicated table, as CSV and Markdown
`slides/`	These slides, in LaTeX



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